Communication Systems Lecture-18: Angle Modulation

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Angle Modulation (1)

- Phase Modulation (PM) and Frequency Modulation (FM) are special cases of angle modulation.
- For this kind of signaling, the complex envelope is given by:

$$g(t) = A_c e^{j\theta(t)}$$

  - A_c is a positive real number.
  - The real envelope is given by $R(t) = |g(t)| = A_c$ is a constant.
  - $g(t)$ is a nonlinear function of the modulating signal $m(t)$.
- The angle-modulated signal can be written as:

$$\begin{aligned} s(t) &= \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\} \\ &= \mathrm{Re}\left\{A_c e^{j\theta(t)} e^{j\omega_c t}\right\} \\ &= A_c \cos\left[\omega_c t + \theta(t)\right] \end{aligned}$$

Angle Modulation (2)

- For PM, the phase is proportional to the modulating signal:

$$\theta(t) = D_p m(t)$$

  where D_p is the phase sensitivity of the PM.
- For FM:

$$\theta(t) = D_f \int_{-\infty}^{t} m(\tau)\mathrm{d}\tau$$

  where D_f is the frequency deviation constant.
- Given that $s(t) = A_c \cos[\omega_c t + \theta(t)]$, the instantaneous frequency of angle-modulated signals is given by:

$$\begin{aligned} f_i(t) &= \frac{1}{2\pi}\frac{\mathrm{d}}{\mathrm{d}t}[\omega_c t + \theta(t)] \\ &= f_c + \frac{1}{2\pi}\left[\frac{\mathrm{d}\theta(t)}{\mathrm{d}t}\right] \end{aligned}$$

- The frequency deviation from the carrier frequency is:

$$f_d(t) = f_i(t) - f_c = \frac{1}{2\pi}\left[\frac{\mathrm{d}\theta(t)}{\mathrm{d}t}\right]$$

Angle Modulation (3)

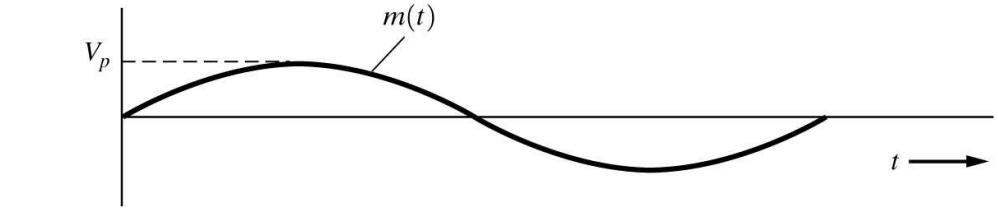
- For FM:

$$f_i(t) = f_c + \frac{1}{2\pi} D_f m(t)$$

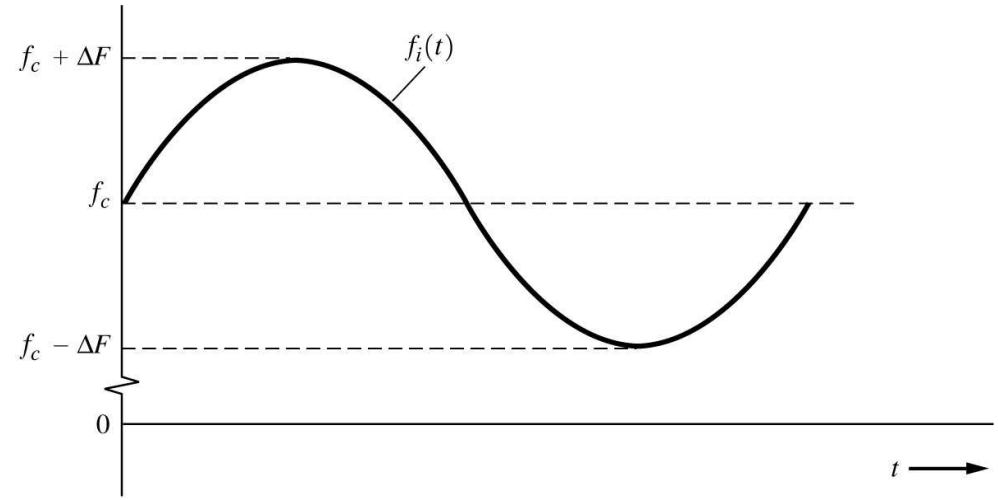
$$f_d(t) = \frac{1}{2\pi} D_f m(t)$$

- This is the reason for calling this type of signaling frequency modulation: the instantaneous frequency varies about the carrier frequency f_c in a manner that is directly proportional to the modulating signal $m(t)$.
- The next figure shows a FM signal when the modulating signal $m(t)$ is sinusoidal.

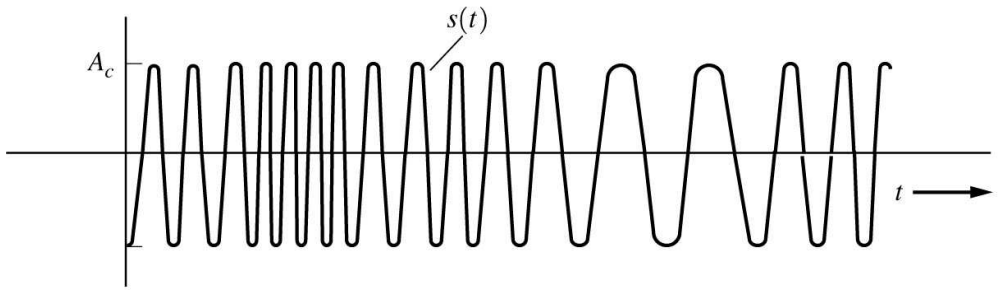
Angle Modulation (4)



(a) Sinusoidal Modulating Signal



(b) Instantaneous Frequency of the Corresponding FM Signal



(c) Corresponding FM Signal

Angle Modulation (5)

- The peak phase deviation is given by:

$$\Delta\theta = \max\left[\theta(t)\right]$$

  - For PM:

$$\Delta\theta = D_p V_p$$

    where $V_p = \max\left[m(t)\right]$.

- The phase modulation index is given by:

$$\beta_p = \Delta\theta$$

Angle Modulation (6)

- The peak frequency deviation is given by:

$$\Delta F = \max\left\{\frac{1}{2\pi}\left[\frac{\mathrm{d}\theta(t)}{\mathrm{d}t}\right]\right\}$$

  - For FM:

$$\Delta F = \frac{1}{2\pi} D_f V_p$$

    where $V_p = \max[m(t)]$.

- The frequency modulation index is given by:

$$\beta_f = \frac{\Delta F}{B}$$

where B is the bandwidth of the modulating signal $m(t)$.

Spectra of Angle-Modulated Signals (1)

- The spectrum of an angle-modulated signal is given by:

$$S(f) = \frac{1}{2}\left[G(f - f_c) + G^*(-f - f_c)\right]$$

where:

$$G(f) = \mathcal{F}\left[g(t)\right] = \mathcal{F}\left[A_c e^{j\theta(t)}\right]$$

- Given that $g(t)$ is a nonlinear function of the modulating signal $m(t)$, then a general formula relating $G(f)$ (and consequently $S(f)$) to $M(f)$ can not be obtained.
- In what follows, we consider the case where $\theta(t)$ takes the following form:

$$\theta(t) = \beta \sin\left[\omega_m t\right]$$

where β is the modulation index.

  - $\beta = \beta_p$ for PM and $\beta = \beta_f$ for FM.

Spectra of Angle-Modulated Signals (2)

- For PM, $\theta(t) = \beta \sin[\omega_m t]$ when the PM signal takes the following form:

$$m_p(t) = A_m \sin[\omega_m t]$$

- In this case:

$$\theta(t) = D_p m_p(t) = D_p A_m \sin[\omega_m t]$$

  implying that $\beta = D_p A_m$.
- On the other hand, the phase modulation index is given by:

$$\beta_p = D_p \max[m_p(t)] = D_p A_m$$

- The last equations imply that:

$$\beta = \beta_p = D_p A_m$$

Spectra of Angle-Modulated Signals (3)

- For FM, $\theta(t) = \beta \sin[\omega_m t]$ when the FM signal takes the following form:

$$m_f(t) = A_m \cos[\omega_m t]$$

- In this case:

$$\frac{\mathrm{d}\theta(t)}{\mathrm{d}t} = D_f m_f(t)$$

$$\Rightarrow \quad \beta\omega_m \cos[\omega_m t] = D_f A_m \cos[\omega_m t]$$

  implying that $\beta = D_f A_m/\omega_m$.
- On the other hand, the frequency modulation index is:

$$\beta_f \triangleq \frac{\Delta F}{B} = \left[\frac{1}{f_m}\right]\left[\frac{1}{2\pi} D_f \max[m_f(t)]\right] = \frac{D_f A_m}{\omega_m}$$

- The last equations imply that:

$$\beta = \beta_f = \frac{D_f A_m}{\omega_m}$$

Spectra of Angle-Modulated Signals (4)

- In both cases, the complex envelope is given by:

$$g(t) = A_c e^{j\theta(t)} = A_c e^{j\beta \sin(\omega_m t)}$$

- $g(t)$ is periodic with period $T_m = 1/f_m$. Consequently, $g(t)$ admits the following Fourier series representation:

$$g(t) = \sum_{n=-\infty}^{+\infty} c_n e^{jn\omega_m t}$$

where:

$$c_n = \frac{1}{T_m} \int_{-\frac{T_m}{2}}^{\frac{T_m}{2}} g(t) e^{-jn\omega_m t} \mathrm{d}t$$

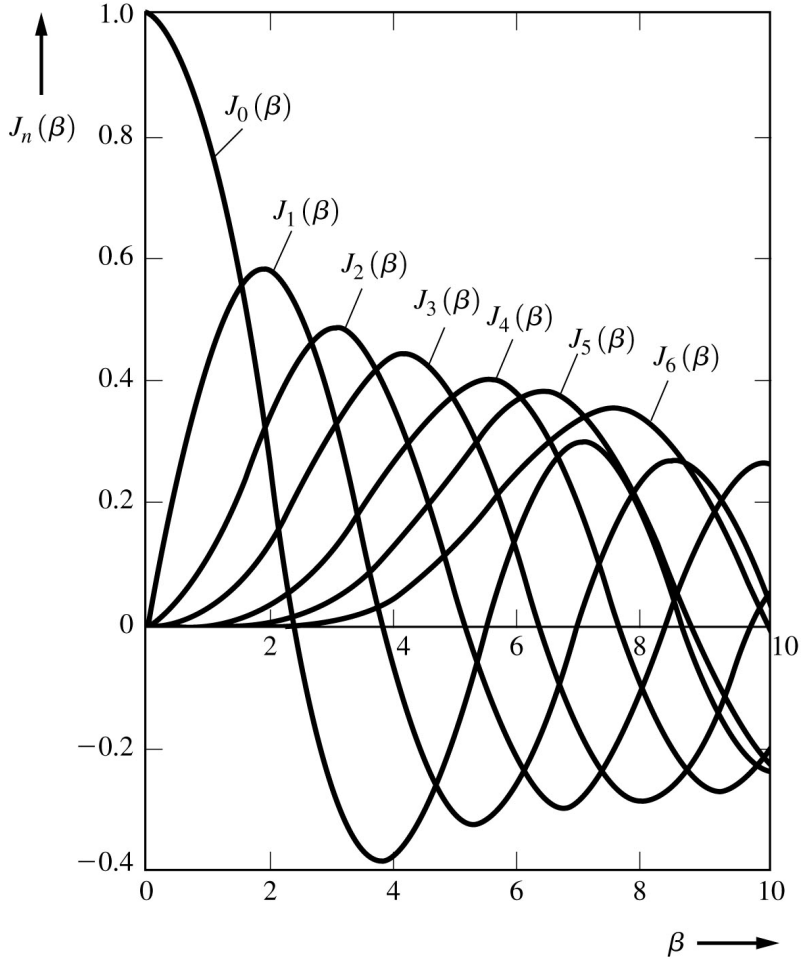
$$= \frac{A_c}{T_m} \int_{-\frac{T_m}{2}}^{\frac{T_m}{2}} e^{j[\beta \sin(\omega_m t) - n\omega_m t]} \mathrm{d}t$$

Spectra of Angle-Modulated Signals (5)

- Let $\theta = \omega_m t$, the last equation reduces to:

$$c_n = A_c \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j[\beta \sin\theta - n\theta]} \mathrm{d}\theta \right]$$
$$\triangleq A_c J_n(\beta)$$

- $J_n(\beta)$ is known as the Bessel function of the first kind of order n and argument β.
- $J_n(\beta)$ can not be evaluated in closed form but it has been evaluated numerically.



Spectra of Angle-Modulated Signals (6)

- It can be shown that;

$$J_{-n}(\beta) = (-1)^n J_n(\beta)$$

  implying that: $|J_{-n}(\beta)| = |J_n(\beta)|$.
- The Fourier transform of the complex envelope is:

$$G(f) = A_c \sum_{n=-\infty}^{+\infty} J_n(\beta)\delta\left(f - nf_m\right)$$

- The spectrum of the angle-modulated signal is:

$$S(f) = \frac{1}{2}\left[G(f - f_c) + G^*(-f - f_c)\right]$$

Spectra of Angle-Modulated Signals (7)

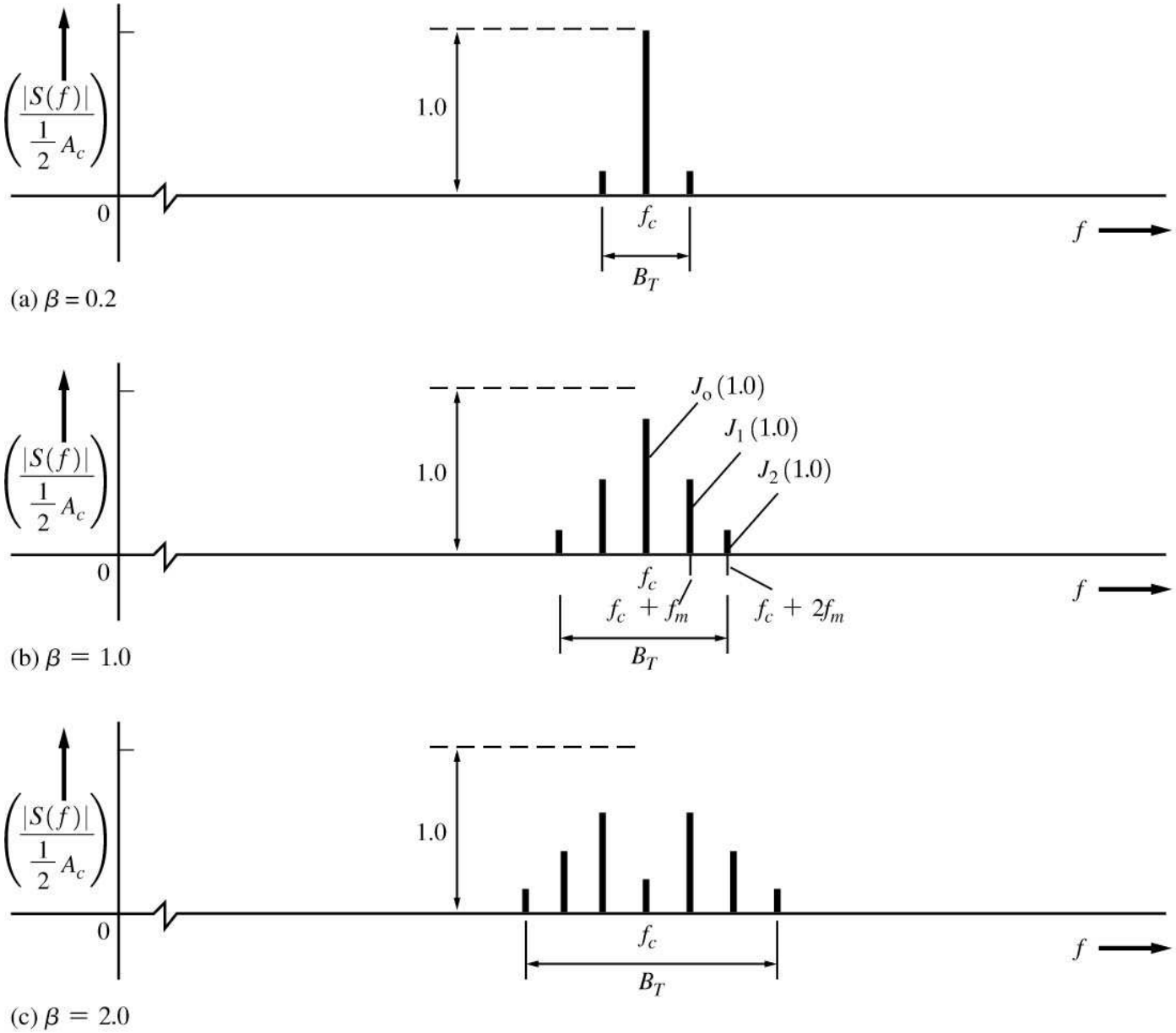
- For $f \geq 0$:

$$|S(f)| = \frac{1}{2}A_c \sum_{n=-\infty}^{+\infty} |J_n(\beta)| \delta\left(f - f_c - nf_m\right)$$

$$= \frac{1}{2}A_c |J_0(\beta)| \delta\left(f - f_c\right) + \frac{1}{2}A_c \sum_{n=1}^{+\infty} |J_n(\beta)| \left\{\delta\left[f - (f_c + nf_m)\right] + \delta\left[f - (f_c - nf_m)\right]\right\}$$

- The magnitude spectrum is symmetrical about the carrier frequency f_c.
- The next figure shows $|S(f)| / (\frac{1}{2}A_c)$ for different values of β.

Spectra of Angle-Modulated Signals (8)



Spectra of Angle-Modulated Signals (9)

- Note that as β increases, new spectral components will appear on both sides of the carrier frequency. Moreover, the spectral components near the carrier frequency will decrease in magnitude since the total power of the signal remains constant.
- **Carson's rule** is a rule-of-thumb for evaluating the bandwidth of PM and FM signals. 98% of the total power is contained in the bandwidth:

$$B_T = 2(\beta + 1)B$$

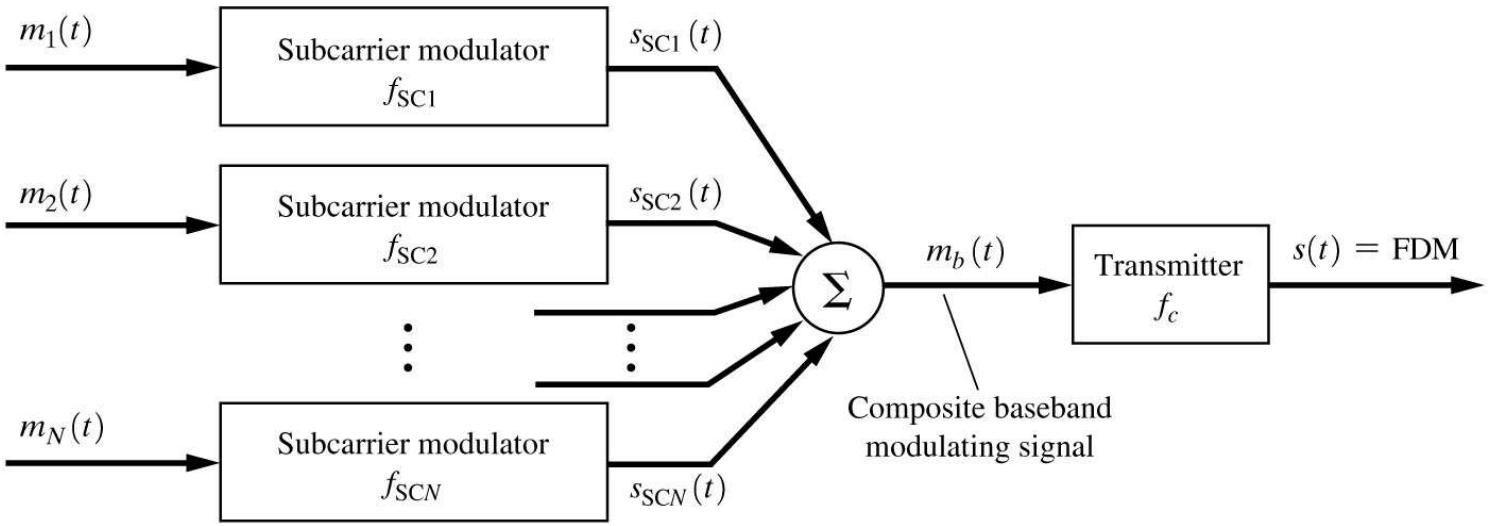
  - β is either the phase modulation index or the frequency modulation index.
  - B is the bandwidth of the modulating signal ($B = f_m$ when $m(t)$ is a sinusoidal function).

Frequency Division Multiplexing (1)

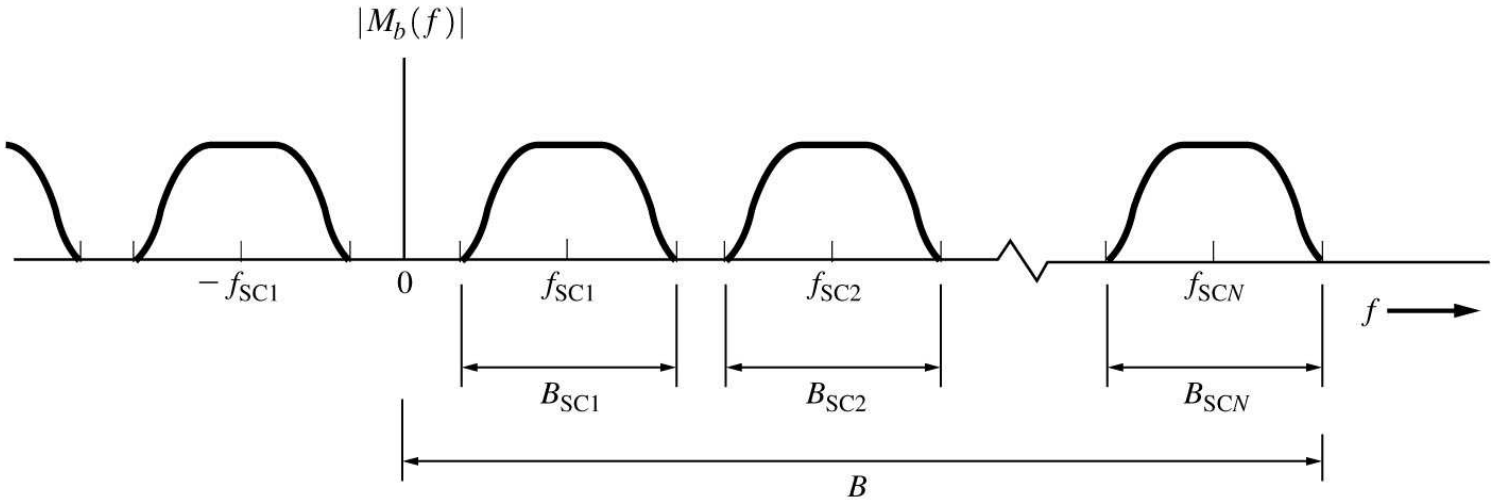
Frequency Division Multiplexing (FDM) is a technique for transmitting multiple messages simultaneously over a wideband channel.

- The multiple messages are first modulated over several subcarriers.
- The result is a composite baseband signal that consists of the sum of these modulated subcarriers. The spectra of the modulated signals must not overlap, otherwise, crosstalk will occur between the different messages.
- The composite baseband signal then modulates the main carrier to produce a passband FDM signal that is transmitted over the wideband channel.
- Note that different modulations can be used over the subcarriers and the main carrier.

Frequency Division Multiplexing (2)



(a) Transmitter

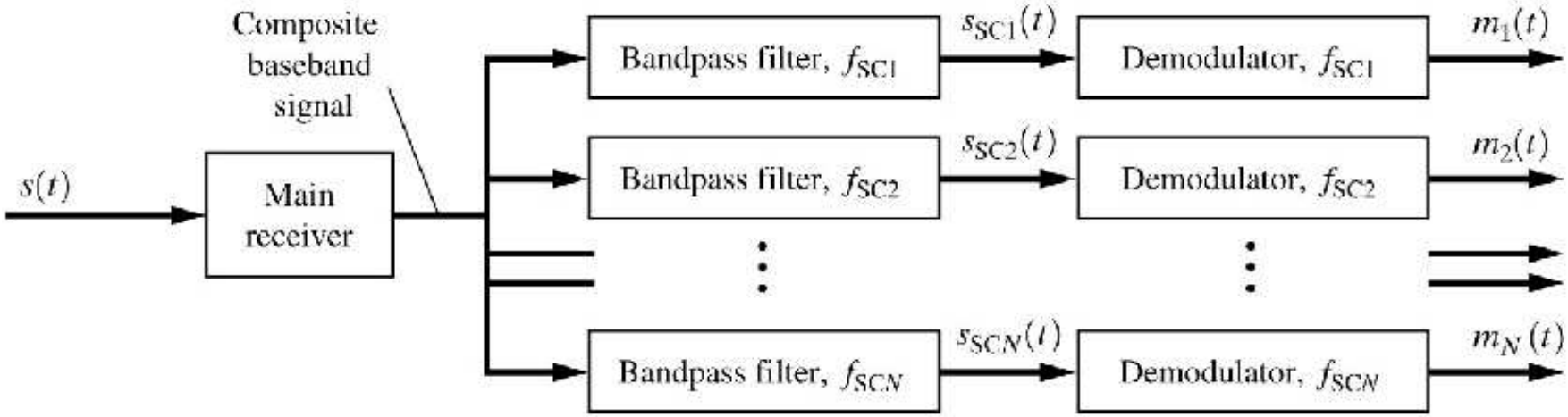


(b) Spectrum of Composite Baseband Signal

Frequency Division Multiplexing (3)

At the receiver:

- The received FDM signal is first demodulated to reproduce the composite baseband signal.
- Filters and demodulators are then used to separate and demodulate the different subcarriers to reproduce the message signals $m_1(t), m_2(t) \ldots$



FM stereo (1)

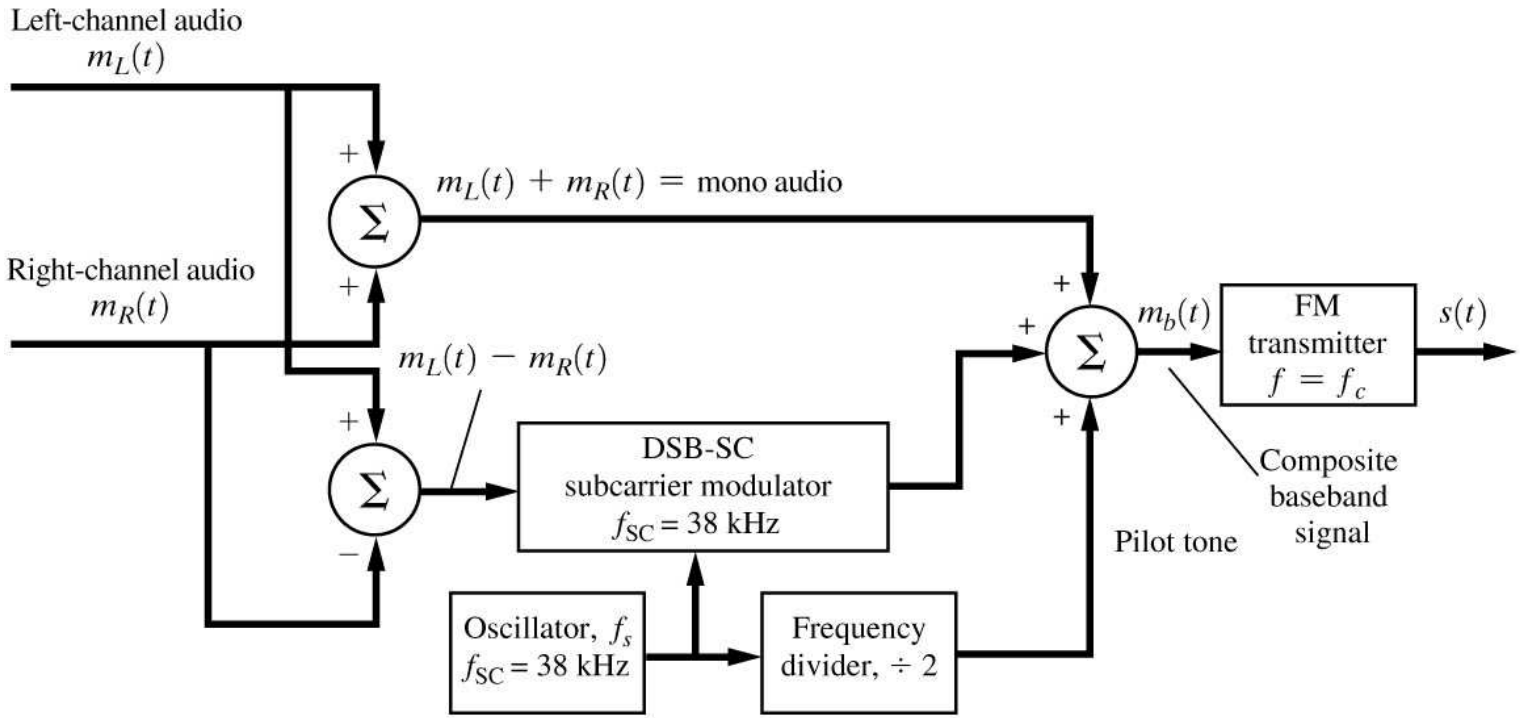
- An example of FDM systems is the FM stereo broadcasting systems.
- The audio signal consists of a right-channel audio and a left-channel audio.
- A listener with a FM stereo receiver receives the right-channel audio on the right speaker and the left-channel audio on the left speaker.
- The FM stereo system must be compatible with monaural FM receivers. In other words, a listener with a monaural FM receiver will listen to the audio signal (left-channel audio plus right-channel audio) from both speakers.

FM stereo (2)

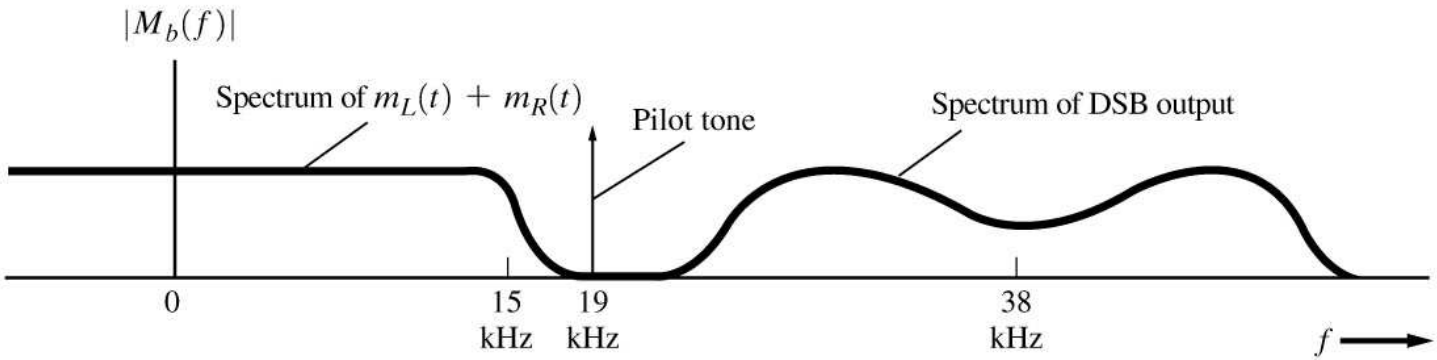
Designate by $m_R(t)$ and $m_L(t)$ the right-channel audio and the left-channel audio respectively.
In order to achieve stereo-monaural compatibility, the composite baseband signal is composed of:

- The sum of the left-channel audio and the right-channel audio $m_L(t) + m_R(t)$. Note that no subcarrier is used for this signal.
- $m_L(t) - m_R(t)$ that will modulate a 38 kHz subcarrier. Double sideband suppressed-carrier (DSB-SC) modulation is used.
- A 19-kHz pilot tone that provides a reference signal for the coherent product detector at the receiver.

FM stereo (3)



(a) FM Stereo Transmitter



(b) Spectrum of Composite Baseband Signal

FM stereo (4)

- The next figure shows that the FM stereo broadcasting systems are compatible with monaural receivers.

